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TRANSFORMABLE SEAT

Abstract

Assemblies, components, and methods are presented herein for a transformable seat. The seat includes a seat back. The seat back incorporates a frame and frame is rotatably coupled at one end to a seat base. The frame at least partially defines an opening in the seat back and two panels are each rotatably coupled to respective sides of the frame.

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Background/Summary

INTRODUCTION

[0001] The present disclosure is directed to a seat that is capable of multiples uses and, more

particularly, to a seat back of a passenger vehicle that is transformable into multiple configurations.
SUMMARY

[0002] There are a number of passenger seat configurations for vehicles. Certain known structures enable a middle seat back of a three passenger seat configuration to be turned into either an arm rest or a through opening. However, these configurations lack a means for providing a modifiable, or customizable, configurations by changing positions of a pair of panels that are integrated into a seat back.

[0003] The seat back of the present disclosure is formed out of a frame. The frame has a pair of panels rotatably coupled to sides of the frame. When the frame of the seat back is arranged horizontal to a seat base, the pair of panels can be rotated about the lateral sides of the frame to expose an opening (e.g., an opening to a storage volume) in the seat back, and the panels may comprise a padded surface to provide proper armrests to passengers seated on either lateral side of the seat back. In some embodiments, the panels can be adjusted to create a work surface.

[0004] In some embodiments, the frame is rotatably coupled to the seat base by a hinge assembly. The hinge assembly is configured to enable the seat back to rotate between an upright position and a folded position where the frame is arranged horizontally above the seat base.

[0005] In some embodiments, left and right detent features are arranged along respective sides of the frame for respective left and right panels (e.g., for ensuring the left and right panels remain in a fixed position once rotated to a desired configuration). When the left and right panels are in a closed position, the left and right panels cover the opening. When the seat back is arranged in an upright orientation and when the left and right panels are rotated outward about respective sides of the frame, the opening is exposed.

[0006] In some embodiments, when the seat back is in a folded down position and when the left and right panels are rotated outward to a first position, the left and right panels form respective armrests. Additionally, or alternatively, when the seat back is in a folded down position and when the left and right panels are rotated outward, the opening is capable of receiving a storage bin that is removably coupled to a perimeter of the opening. In some embodiments, when the seat back is in a folded down position and when the left and right panels are rotated outward to a second position, the left and right panels are horizontally arranged and form respective work surfaces.

[0007] In some embodiments, the seat back also includes a head rest rotatably coupled to a second end of the seat back arranged opposite the one end rotatably coupled to the seat base. Additionally, or alternatively, the frame is comprised of a plurality of tubular members fixedly attached to ends of each other.

[0008] In some embodiments, the disclosure is directed to a vehicle assembly. The vehicle assembly includes a vehicle cabin and a row of seats arranged within the vehicle cabin, the row of passenger seats comprise at least one seat back of this disclosure. In some embodiments, the row of seats comprises three seats. Additionally, or alternatively, the at least one seat back comprises a seat belt retractor coupled to the frame.

[0009] In some embodiments, the disclosure is directed to a method of transforming a seat back. A top end of a frame of the seat back is rotatably articulated from an upright orientation to a folded down orientation via a hinge assembly at a bottom end of the frame. Left and right panels are rotated about respective sides of the frame to expose an opening at least partially defined by the frame. Additionally, or alternatively, the method includes arranging a storage bin within the opening.

Description

BRIEF DESCRIPTIONS OF THE DRAWINGS

[0010] The above and other objects and advantages of the disclosure may be apparent upon

consideration of the following detailed description, taken in conjunction with the accompanying drawings, in which:

- [0011] FIG. 1A illustrates a vehicle cabin with a row of seats including an exemplary seat back, in accordance with some embodiments of the disclosure;
- [0012] FIG. 1B illustrates a vehicle cabin with a row of seats including an exemplary seat in a folded configuration, in accordance with some embodiments of the disclosure;
- [0013] FIG. 1C illustrates a vehicle cabin with a row of seats including an exemplary seat back with access to an opening, in accordance with some embodiments of the disclosure;
- [0014] FIG. 1D illustrates a vehicle cabin with a row of seats including an exemplary seat back manipulated to create a platform, in accordance with some embodiments of the disclosure;
- [0015] FIG. 2A illustrates a side view of an exemplary seat in a folded configuration, in accordance with some embodiments of the disclosure;
- [0016] FIG. 2B illustrates a cross sectional view of the seat of FIG. 2A in folded configuration, in accordance with some embodiments of the disclosure;
- [0017] FIG. 2C illustrates a cross sectional view of the seat of FIG. 2A in a folded configuration with access to an opening, in accordance with some embodiments of the disclosure;
- [0018] FIG. 2D illustrates a cross sectional view of the seat of FIG. 2A in a folded configuration where the seat back is manipulated to create a platform, in accordance with some embodiments of the disclosure;
- [0019] FIG. 2E illustrates a cross sectional view of the seat of FIG. 2A in a folded configuration with access to an opening, in accordance with some embodiments of the disclosure;
- [0020] FIG. 3 illustrates a seat with a seat back in an upright position and the seat back being manipulated such that the seat is in a folded configuration, in accordance with some embodiments of the disclosure;
- [0021] FIG. 4 illustrates a rear view of a seat assembly of this disclosure, in accordance with some embodiments of the disclosure;
- [0022] FIG. 5 illustrates a seat assembly with a temperature-controlled storage area, in accordance with some embodiments of the disclosure;
- [0023] FIG. 6A illustrates a seat assembly with a removable storage area, in accordance with some embodiments of the disclosure;
- [0024] FIG. 6B illustrates the seat assembly of FIG. 6A with the storage area pulled out of a base of the seat assembly, in accordance with some embodiments of the disclosure;
- [0025] FIG. 6C illustrates the seat assembly of FIG. 6A in a folded configuration with a removable storage area, in accordance with some embodiments of the disclosure;
- [0026] FIG. 6D illustrates the seat assembly of FIG. 6A in the folded configuration with the storage area pulled out of the base of the seat assembly, in accordance with some embodiments of the disclosure;
- [0027] FIG. 7A illustrates a seat assembly with a seat back that has a padded panel interfacing with a seat back frame, in accordance with some embodiments of the disclosure;
- [0028] FIG. 7B illustrates the seat assembly of FIG. 7A where the seat back is being manipulated to separate a paddle panel from a seat back frame, in accordance with some embodiments of the disclosure;
- [0029] FIG. 7C illustrates a platform configuration of the seat assembly of FIG. 7A, in accordance with some embodiments of the disclosure;
- [0030] FIG. 8 illustrates a frame of a seat back of the disclosure and a removeable storage bin, in accordance with some embodiments of the disclosure;
- [0031] FIG. 9 is a flow chart representing an illustrative process for manipulating a transformable seatback of the disclosure, in accordance with some embodiments of the disclosure.

DETAILED DESCRIPTION

[0032] Methods, components, and assemblies are provided herein for a transformable seat. The

transformable seat includes a seat back of a vehicle with a pair of rotatable panels which provide access to different seat configurations depending on the arrangement of the pair of rotatable panels relative to the seat back.

[0033] FIG. 1A depicts vehicle cabin **100A** of a vehicle assembly with row of seats **102** comprising seat back **104**, in accordance with some embodiments of the disclosure. Vehicle cabin **100A** may include or incorporate any or all elements depicted in, or described in reference to, FIGS. 1B-8. One or more elements of vehicle cabin **100A** may be manipulated or adjusted based on one or more elements of process **900** of FIG. 9.

[0034] In some embodiments, vehicle cabin **100A** is considered a passenger or occupant portion of a vehicle (e.g., a vehicle assembly). For example, steering column assembly **114** is arranged in front of an end seat of row of seats **102** to enable an occupant of a vehicle comprising vehicle cabin **100A** to steer the vehicle. Secured, or adjustably secured, to cabin floor **106** is row of seats **102**. Row of seats **102** as shown in FIGS. 1A-ID comprises three passenger seats. Row of seats **102** may comprise any of a pair of seats, a plurality of seats, or a single seat. Row of seats **102** may be considered one or more of a front row of passenger seats, a middle row of passenger seats, or a rear row of passenger seats. A middle seat of the three passenger seats includes seat back **104**. In some embodiments, seat back **104** is incorporated into either or both of the two end seats. In some embodiments, seat back **104** is incorporated into one or more of the passenger seats of row of seats **102**. Seat back **104** comprises frame **108**, which at least partially defines an opening in seat back **104**. The opening is covered in FIG. 1A by left panel **110A** and right panel **110B** as the panels are arranged in FIG. 1A in a closed position relative to seat back **104**. Left panel **110A** is rotatably coupled to a left side of frame **108**. Right panel **110B** is rotatably coupled to a right side of frame **108**. Frame **108** is rotatably coupled at one end to seat base **112**. For example, a hinge assembly with multiple detent features for multiple angles may be incorporated into the middle seat of FIG. 1A to enable seat back **104** to be articulated from an upright position to a folded position along seat base **112**. In some embodiments, frame **108** is rotatably coupled to seat base **112** by one or more of an actuator or a radial spring mechanism. Left panel **110A** and right panel **110B** can be rotated outwards around sides of frame **108** to expose an opening when seat back **104** is arranged in the upright position of FIG. 1A.

[0035] FIG. 1B depicts vehicle cabin **100B** with seat back **104** arranged in a folded configuration such that seat back **104** extends along seat base **112**, in accordance with some embodiments of the disclosure. Vehicle cabin **100B** may include or incorporate any or all elements depicted in, or described in reference to, FIGS. 1A, and 1C-8. One or more elements of vehicle cabin **100B** may be manipulated or adjusted based on one or more elements of process **900** of FIG. 9.

[0036] Vehicle cabin **100B** incorporates all the elements shown in and described in reference to FIG. 1A. Frame **108**, as described in reference to FIG. 1A, is rotatably coupled to seat base **112** such that seat back **104** articulates, or rotates about a hinge assembly, between the upright position shown in FIG. 1A and the folded position depicted in FIG. 1B. The folded position of seat back **104** includes frame **108** being arranged horizontally above seat base **112**. Frame **108** also defines opening **116**. Opening **116** is covered on one side of frame **108** by left panel **110A** and right panel **110B**. As shown in FIG. 1B, frame **108** forms a recess between rear surfaces of left panel **110A** and right panel **110B** when seat back **104** is arranged in the folded position.

[0037] FIG. 1C depicts vehicle cabin **100C** where seat back **104** is in the folded position with opening **116** extending all the way through seat back **104** based on the position of left panel **110A** and right panel **110B**, in accordance with some embodiments of the disclosure. Vehicle cabin **100C** may include or incorporate any or all elements depicted in, or described in reference to, FIGS. 1A, 1B, and 1D-8. One or more elements of vehicle cabin **100C** may be manipulated or adjusted based on one or more elements of process **900** of FIG. 9.

[0038] Vehicle cabin **100C** incorporates all the elements shown in and described in reference to FIGS. 1A and 1B. Seat back **104** is in the folded position described in reference to FIG. 1B. Left

panel **110A** and right panel **110B** are rotated outwards about sides of frame **108**. As a result, opening **116** provides access all the way through seat back **104** such that access to a storage area is provided to an occupant of vehicle cabin **100C**. In some embodiments, the storage area ends at the top surface of seat base **112**. In some embodiments, the storage area may be extended to cabin floor **106** based on manipulation of seat base **112** (e.g., removing or articulating a seat cushion to expose more space for storing items through opening **116**). As shown in FIG. **1C**, left panel **110A** has an elevated padded structure in this depicted orientation. This elevated padded structure forms a first armrest. Left panel **110B** also has an elevated padded structure in the depicted orientation. This second elevated padded structure forms a second armrest arranged on a side of frame **108** opposite the first armrest. In some embodiments, opening **116** is configured to receive a storage bin that is removably coupled to a perimeter of opening **116** (e.g., as at least partially defined by frame **108**). [0039] FIG. **1D** depicts vehicle cabin **100D** where seat back **104** is manipulated to create platform **118**, in accordance with some embodiments of the disclosure. Vehicle cabin **100D** may include or incorporate any or all elements depicted in, or described in reference to, FIGS. **1A-1C** and **2A-8**. One or more elements of vehicle cabin **100D** may be manipulated or adjusted based on one or more elements of process **900** of FIG. **9**.

[0040] Vehicle cabin **100D** incorporates all the elements shown in and described in reference to FIGS. **1A-1C**. Seat back **104** is in the folded position described in reference to FIGS. **1B** and **1C**. Left panel **110A** and right panel **110B** are rotated outwards about sides of frame **108** beyond the rotation required to result in the configuration of FIG. **1C**. As a result, surfaces of left panel **110A** and right panel **110B** meet over opening **116** to create platform **118**. In some embodiments, platform **118** is not a single work surface formed by ends of surfaces of left panel **110A** and right panel **110B** contacting over opening **116**. As a result, platform **118** may comprise two separate work surfaces when respective portions of left panel **110A** and right panel **110B** are arranged horizontally (e.g., with respect to seat base **112**). For example, the respective work surfaces may be adjusted towards or away from each other to remove or provide access to opening **116**, and a storage volume accessible via opening **116**.

[0041] FIG. **2A** depicts side view **200A** of seat assembly **202** in a folded configuration, in accordance with some embodiments of the disclosure. Seat assembly **202** may include or incorporate any or all elements depicted in, or described in reference to, FIGS. **1A-1D** and **2B-8**. One or more elements of seat assembly **202** may be manipulated or adjusted based on one or more elements of process **900** of FIG. **9**.

[0042] Seat assembly **202** includes seat base **204**. Seat back **206** is rotatably coupled to seat base **204** via hinge assembly **208**. Hinge assembly **208** comprises any suitable apparatus or assembly for enabling seat back **206** to articulate between an upright position (e.g., suitable for accommodating a seat vehicle occupant on seat base **204**) and the horizontal or folded position depicted in FIG. **2A**. Seat back **206** includes a pair of panels rotatably coupled to sides of a frame, which are each shown in more detail in FIGS. **2B-2E**. Arranged below seat base **204** is storage component **210**. Storage component **210** may be a tray or other removeable volume for accommodating one or more devices or objects of a vehicle occupant. For example, storage component **210** may be removably attached or slidably attached to seat frame **212**. Seat frame **212** corresponds to any component, assembly, subassembly, or interface that enables seat assembly **202** to be secured, or adjustably secured, to a surface of a floor of a vehicle cabin (e.g., cabin floor **106** of FIG. **1A**). Storage component **210** is divided into storage areas **214A** and **214B** by storage divider **216**. Storage divider **216** may comprise any suitable structural configuration to prevent one or more of objects, devices, or components stored in storage areas **214A** and **214B** from contacting each other while a vehicle assembly comprised of seat assembly **202** is moving.

[0043] FIG. **2B** depicts cross sectional view **200B** of seat assembly **202** in a folded configuration, in accordance with some embodiments of the disclosure. Seat assembly **202** may include or incorporate any or all elements depicted in, or described in reference to, FIGS. **1A-2A** and **2C-8**.

One or more elements of seat assembly **202** may be manipulated or adjusted based on one or more elements of process **900** of FIG. **9**.

[0044] Cross sectional view **200B** depicts seat assembly **202** along cross section A-A from FIG. **2A**. Seat back **206** is comprised of left panel **110A** and right panel **110B**. Left panel **110A** and right panel **110B** are rotatably coupled to frame **218** (e.g., corresponding to frame **108** of FIGS. **1A-1D**). In some embodiments, frame **218** comprises one or more detent features on each lateral side to enable articulation of left panel **110A** and right panel **110B** into the varying positions of this disclosure (e.g., as shown in and described in reference to FIGS. **1A-1D**). For example, detent features **220A** and **220B** are shown as interfacing with both frame **218** and respective panels. Detent feature **220A** enables left panel **110A** to rotatably articulate about a first side of frame **218** as shown in FIGS. **2B-2E** for different configurations, or arrangements, corresponding to different secured, or locked, positions of left panel **110A** relative to the first side of frame **218**. Detent features **220B** enable right panel **110B** to rotatably articulate about a second side of frame **218** as shown in FIGS. **2B-2E** for different configurations, or arrangements, corresponding to different secured, or locked, positions of right panel **110B** relative to the second side of frame **218**. Detent features **220A** and **200B** may be one or more of an extension, a recess, a locking feature, a releasing feature, or interface that enables each of left panel **110A** and right panel **110B** to be articulated, or manipulated, around respective sides of frame **218**.

[0045] FIG. **2C** depicts cross sectional view **200C** of seat assembly **202** in a folded configuration with access to opening **116**, in accordance with some embodiments of the disclosure. Seat assembly **202** may include or incorporate any or all elements depicted in, or described in reference to, FIGS. **1A-2B** and **2D-8**. One or more elements of seat assembly **202** may be manipulated or adjusted based on one or more elements of process **900** of FIG. **9**.

[0046] Cross sectional view **200C** depicts seat assembly **202** along cross section A-A from FIG. **2A**. As shown in FIG. **2C**, detent feature **220A** is used to arrange left panel **110A** such that opening **232** of frame **218** is at least partially accessible or exposed. Detent feature **220B** is used to arrange right panel **110B** in a similar fashion relative to frame **218** to at least partially expose, or create access to, opening **232**. Arranged within opening **232** is storage volume **222**. In some embodiments, storage volume **222** is removably attached (e.g., at one or more of a flange, a lip, or one or more extensions) to a perimeter of frame **218**. In some embodiments, storage volume **222** is a collapsible structure. For example, storage volume **222** may be part of, or configured to be arranged within, storage compartment **210** prior to arrangement within opening **232**. Detent features **220A** and **220B** may be the same features used in FIG. **2A** to achieve the arrangement of left panel **110A** and right panel **110B**. In some embodiments, the detent features of FIG. **2A** may be different than the detent features of FIG. **2B**. Panel surfaces **226A** and **226B** can be considered armrests in FIG. **2C**. Additionally, or alternatively, padding may be integrated into panel surfaces **226A** and **226B** for comfort of occupants of a vehicle comprising seat assembly **202**.

[0047] FIG. **2D** depicts cross sectional view **200D** of seat assembly **202** in a folded configuration where left panel **110A** and right panel **110B** are manipulated to create work surfaces **224A** and **224B** that, when pushed together, can form a platform, in accordance with some embodiments of the disclosure. Seat assembly **202** may include or incorporate any or all elements depicted in, or described in reference to, FIGS. **1A-2C** and **2E-8**. One or more elements of seat assembly **202** may be manipulated or adjusted based on one or more elements of process **900** of FIG. **9**.

[0048] Cross sectional view **200D** depicts seat assembly **202** along cross section A-A from FIG. **2A**. As shown in FIG. **2D**, detent feature **234A** is used to arrange left panel **110A** such that opening **232** of frame **218** is at least partially accessible or exposed. Detent feature **234B** is used to arrange right panel **110B** in a similar fashion relative to frame **218** to at least partially expose, or create access to, opening **232**. Arranged within opening **232** is storage volume **222**. In some embodiments, storage volume **222** is removably attached (e.g., at one or more of a flange, a lip, or one or more extensions) to a perimeter of frame **218**. In some embodiments, one or more of detent

features **234A** and **234B** may be used to arrange storage volume **222** to be removably coupled to frame **218** at a perimeter of frame **218**. When left panel **110A** and right panel **110B** are arranged as shown in FIG. **2D**, work surfaces **224A** and **224B** are accessible. In some embodiments, work surfaces **224A** and **224B** can be pushed together such that panel surfaces **226A** and **226B** are touching. Accordingly, a platform may be formed (e.g., platform **118**). As shown in FIG. **2D**, panel surfaces **236A** and **236B** extend below a top surface of frame **218** when left panel **110A** and **110B** are manipulated, or articulated, to expose work surfaces **224A** and **224B**. In some embodiments, panel surfaces **236A** and **236B** may be arranged above a top surface of frame **218**.

[0049] FIG. **2E** illustrates a cross sectional view of the seat of FIG. **2A** in a folded configuration with access to an opening, in accordance with some embodiments of the disclosure. Seat assembly **202** may include or incorporate any or all elements depicted in, or described in reference to, FIGS. **1A-2D** and **3-8**. One or more elements of seat assembly **202** may be manipulated or adjusted based on one or more elements of process **900** of FIG. **9**.

[0050] Cross sectional view **200E** depicts seat assembly **202** along cross section A-A from FIG. **2A**. As shown in FIG. **2E**, a seat back comprising frame **218**, left panel **110A**, and right panel **110B** is arranged in a folded down position such that frame **218** extends along seat base **204**. Left panel **110A** and right panel **110B** are rotated outward relative to frame **118** such that both of work surfaces **224A** and **224B** are aligned with a top surface of seat base **204**. In this arrangement of panels, opening **232** is fully exposed such that opening **232** is capable of receiving storage volume **222** (e.g., a storage bin or collapsible storage volume). As described in reference to FIGS. **2C-2E**, storage volume **222** is removably coupled to a perimeter of opening **232** that is part of frame **218**. Panel surfaces **228A** and **228B** may be utilized as arm rests for occupants of a vehicle comprised of seat assembly **202** when left panel **110A** and right panel **110B** are arranged as shown in FIG. **2E**. In some embodiments, panel surfaces **230A** and **230B** may be pushed towards each other until they contact each other to create a smaller volume of storage within frame **218** (e.g., as shown in FIG. **1B**).

[0051] FIG. **3** depicts seat assembly **300** in a first configuration with seat back **302** arranged to align with upright alignment **304** and seat assembly **300** in a second configuration with seat back **302** arranged in to align with folded alignment **306**, in accordance with some embodiments of the disclosure. Seat assembly **300** may include or incorporate any or all elements depicted in, or described in reference to, FIGS. **1A-2E** and **4-8**. One or more elements of seat assembly **300** may be manipulated or adjusted based on one or more elements of process **900** of FIG. **9**.

[0052] Seat assembly **300** is comprised of seat back **302**, which is rotatably coupled to seat base **112** via hinge assembly **208**. Headrest **308** is coupled to top end **310** of seat back **302** and can be rotatably articulated relative to top end **310**. For example, after seat back **302** is articulated about hinge assembly **208** relative to seat base **112** from upright alignment **304** to folded alignment **306** along seat back articulation path **312**, headrest **308** can be manipulated such that headrest top **314** remains above seat back **302**. Headrest **308** is arranged on seat back **302** at top end **310**, which is arranged opposite the end of seatback **302** rotatably coupled via hinge assembly **208** to seat base **112**. Headrest **308** may be manipulated to prevent longer items stored along seat base **112** from sliding. When seat back **302** is aligned with folded alignment **306** (e.g., a folded down position), left panel **110A** and right panel **110B** are shown in FIG. **3** as being rotated outward relative to respective sides of frame **218**. This exposes an opening in which a storage bin can be removably inserted (e.g., at a perimeter of the opening).

[0053] FIG. **4** depicts a rear view of seat assembly **400**, in accordance with some embodiments of the disclosure. Seat assembly **400** may include or incorporate any or all elements depicted in, or described in reference to, FIGS. **1A-3** and **5-8**. One or more elements of seat assembly **400** may be manipulated or adjusted based on one or more elements of process **900** of FIG. **9**.

[0054] Seat assembly **400** comprises seats **402A-C**. Seats **402A** and **402C** incorporate outer seat anchors **404** for securing seats **402A-C** to seat frame **406**. Seat frame **406** is fixedly attached to

cabin floor **106**. Airbags **408** are arranged towards the ends of seat assembly **400**, with an additional airbag **408** arranged between seat **402A** and seat **402B**. Arranged towards a base of each of seats **402A-C** is an occupant classification circuitry, represented by occupant classification circuitries **410A-C**, respectively. Each of occupant classification circuitries comprise one or more sensors for collecting data regarding an occupant of each respective seat. For example, occupant **412** is depicted as sitting in seat **402B**. Accordingly, occupant classification circuitry **410B** is configured to collected data related to occupant **412**. Arranged towards tops of seats **402A** and **402C** are seatbelt retractors **414A** and **414B**. Seatbelt retractor **414A** may be communicatively coupled to occupant classification circuitry **410A** such that seatbelt retractor **414A** is configured to perform locking actions based on one or more characteristics of an occupant of seat **402A**. Seatbelt retractor **414B** may be communicatively coupled to occupant classification circuitry **410C** such that seatbelt retractor **414B** is configured to perform locking actions based on one or more characteristics of an occupant of seat **402C**. Regarding seat **402B** with occupant **412**, seat back **418** is rotatably coupled to seat base **422** via hinge assembly **208**. Arranged above hinge assembly **208** is seatbelt **420**. Seatbelt **420** may have a length adjusted relative to occupant **412** based on one or more of retractor **416A** or retractor **416B**, depending on an expected weight corresponding to occupant **412**. One or more of retractor **416A** or retractor **416B** may be communicatively coupled to occupant classification circuitry **410B** in order to guide length restrictions or adjustments of seatbelt **420** based on length modification accomplished via one or more of retractor **416A** or retractor **416B**. Retractors **416A** and **416B** are coupled to a frame of seatback **418**.

[0055] FIG. 5 depicts seat assembly **500** with temperature-controlled storage area comprised of storage compartment **210**, in accordance with some embodiments of the disclosure. Seat assembly **500** may include or incorporate any or all elements depicted in, or described in reference to, FIGS. **1A-4** and **6A-8**. One or more elements of seat assembly **500** may be manipulated or adjusted based on one or more elements of process **900** of FIG. **9**.

[0056] Scat assembly **500** comprises seat back **206** and seat base **204**. Seat assembly **500** is configured to support occupant **502**. Arranged below seat base **204** is storage compartment **210**. Occupant compartment surface **504** supports storage compartment **210** and seat assembly **500**. Interfacing with occupant compartment surface **504** and storage compartment **210** is temperature control circuitry **506**. Temperature control circuitry **506** is configured to regulate a temperature of one or more of storage compartment **210** and seat base **204**. For example, temperature control circuitry **506** may be configured to modify conditions within storage compartment **210** such that storage compartment **210** maintains an elevated temperature for keeping food or items warm or a reduced temperature for avoiding food from becoming too warm for consumption. Temperature control circuitry **506** may be communicatively coupled to a remote personal device or an occupant interface of a vehicle comprising seat assembly **500**.

[0057] FIG. **6A** depicts seat assembly **600A** with removable storage area comprising storage compartment **210** partially extending beyond a front edge of seat base **204**, in accordance with some embodiments of the disclosure. Seat assembly **600A** may include or incorporate any or all elements depicted in, or described in reference to, FIGS. **1A-5** and **6B-8**. One or more elements of seat assembly **600A** may be manipulated or adjusted based on one or more elements of process **900** of FIG. **9**. Removable storage area **602** corresponds to storage component **210** of FIG. **2A**.

[0058] FIG. **6B** depicts seat assembly **600A** with removable storage area comprising storage compartment **210** pulled out from under seat base **204**, in accordance with some embodiments of the disclosure.

[0059] FIG. **6C** depicts seat assembly **600A** in a folded configuration with removable storage area comprising storage compartment **210** partially extending beyond a front edge of seat base **204**, in accordance with some embodiments of the disclosure. For example, at least one detent feature that extends from or interfaces with occupant compartment surface **504** may be used to secure storage compartment **210** at different extension lengths relative to seat base **204**. Accordingly, a front

portion of storage compartment **210** comprises a cup holder. Seat assembly **600A** achieves the depicted folded configuration as seat back **206** is rotatably coupled to seat base **204** via hinge assembly **208**. In some embodiments, a top of seat back **206** is configured to serve as one or more of an armrest, a platform, one or more work surfaces, or a storage area access threshold.

[0060] FIG. **6D** depicts seat assembly **600A** in the folded configuration with removable storage area pulled out from under seat base **204**, in accordance with some embodiments of the disclosure.

[0061] FIG. **7A** depicts seat assembly **702** in configuration **700A** with seat back **704** that has padded panel **706** interfacing with seat back frame **708**, in accordance with some embodiments of the disclosure. Seat assembly **702** may include or incorporate any or all elements depicted in, or described in reference to, FIGS. **1A-6D** and **7B-8**. One or more elements of seat assembly **702** may be manipulated or adjusted based on one or more elements of process **900** of FIG. **9**. Removable storage area **602** corresponds to storage component **210** of FIG. **2A**.

[0062] Seat back **704** includes hinge assembly **208** to rotatably articulate or manipulate seat back **704** relative to seat base **204**. Additionally, seat back **704** includes panel hinge assembly **710** for rotating padded panel **706** relative to seat back frame **708**. Arranged below seat base **204** is storage component **210**.

[0063] FIG. **7B** depicts configuration **700B** of seat assembly **702** where padded panel **706** is rotated relative to seat back frame **708** via panel hinge assembly **710** away from seat back frame **708**, in accordance with some embodiments of the disclosure. A bottom end of padded panel **706** is pulled away from hinge assembly **208** such that seat back frame **708** is pulled towards a horizontal orientation over seat base **204**.

[0064] FIG. **7C** depicts configuration **700C** of seat assembly **702**, in accordance with some embodiments of the disclosure. Configuration **700C** has a bottom end of padded panel **706** arrange proximate to, or contacting, occupant compartment surface **504**. Additionally, storage component **210** is arranged within seat back frame **708** (e.g., by interfacing with one or more elements of an inner perimeter of seat back frame **708**) after storage component **210** is removed from below seat base **204** and placed between hinge assembly **208** and panel hinge assembly **706**.

[0065] FIG. **8** depicts frame **800** of a seat back of the disclosure and removable storage bin **802**, in accordance with some embodiments of the disclosure. Frame **800** may include or incorporate any or all elements depicted in, or described in reference to, FIGS. **1A-7C**. One or more elements of frame **800** may be manipulated or adjusted based on one or more elements of process **900** of FIG. **9**.

[0066] Frame **800** includes inner perimeter **804** which is configured to interface with sides of removable storage bin **802**. Inner perimeter **804** is exposed when left panel **806A** and right panel **806B** are rotatably articulated away from inner perimeter **804** about left hinge assembly **810A** and right hinge assembly **810B** to expose opening **812**. Opening **812** corresponds to, for example, opening **232** of FIGS. **2C** and **2D**. Arranged along a left side of inner perimeter **804** are left detent features **808A**. Left detent features **808A** may be any suitable construction or configuration to enable one or more of the various panel arrangements described in this disclosure when manipulating, or articulating, left panel **806A**. Arranged along a right side of inner perimeter **804** are right detent features **808B**. Right detent features **808B** may be any suitable construction or configuration to enable one or more of the various panel arrangements described in this disclosure when manipulating, or articulating, right panel **806B**. Although the plurality of members forming frame **800** are depicted as comprising flat surfaces, the plurality of members may also comprise a plurality of tubular members fixedly attached to ends of each other in order to enable smooth rotational articulation of one or more of left panel **806A** or right panel **806B**.

[0067] FIG. **9** is a flow chart representing an illustrative process for manipulating a transformable seatback of the disclosure, in accordance with some embodiments of the disclosure.

[0068] At process block **902**, a top end of a frame of a seat back is rotatably articulated from an upright orientation to a folded down orientation via a hinge assembly at a bottom end of the frame

(e.g., as shown in FIG. 3). At process block **904**, left and right panels are rotated about respective sides of the frame to expose an opening that is at least partially defined by the frame. At process block **906**, a storage bin is arranged within the opening. For example, FIGS. 1A-3 provide various examples of how the panels can be rotated and articulated relative to the frame.

[0069] The systems and processes discussed above are intended to be illustrative and not limiting. One skilled in the art would appreciate that the actions of the processes discussed herein may be omitted, modified, combined, and/or rearranged, and any additional actions may be performed without departing from the scope of the invention. More generally, the above disclosure is meant to be exemplary and not limiting. Only the claims that follow are meant to set bounds as to what the present disclosure includes. Furthermore, it should be noted that the features and limitations described in any one embodiment may be applied to any other embodiment herein, and flowcharts or examples relating to one embodiment may be combined with any other embodiment in a suitable manner, done in different orders, or done in parallel. In addition, the systems, components, and methods described herein may be performed in real time. It should also be noted that the systems, components, and/or methods described above may be applied to, or used in accordance with, other systems, components, and/or methods.

Claims

1. A seat back, comprising: a frame rotatably coupled at one end to a seat base, wherein the frame at least partially defines an opening in the seat back; and left and right panels rotatably coupled to respective sides of the frame.
2. The seat back of claim 1, wherein: the frame is rotatably coupled to the seat base by a hinge assembly; and the hinge assembly is configured to enable the seat back to rotate between an upright position and a folded position where the frame is arranged horizontally above the seat base.
3. The seat back of claim 1, further comprising left and right detent features arranged along respective sides of the frame for respective left and right panels.
4. The seat back of claim 1, wherein when the left and right panels are in a closed position, the left and right panels cover the opening.
5. The seat back of claim 1, wherein when the seat back is arranged in an upright orientation and when the left and right panels are rotated outward about respective sides of the frame, the opening is exposed.
6. The seat back of claim 1, wherein when the seat back is in a folded down position and when the left and right panels are rotated outward to a first position, the left and right panels form respective armrests.
7. The seat back of claim 1, wherein when the seat back is in a folded down position and when the left and right panels are rotated outward, the opening is capable of receiving a storage bin that is removably coupled to a perimeter of the opening.
8. The seat back of claim 1, wherein when the seat back is in a folded down position and when the left and right panels are rotated outward to a second position, the left and right panels are horizontally arranged and form respective work surfaces.
9. The seat back of claim 1, further comprising a head rest rotatably coupled to a second end of the seat back arranged opposite the one end rotatably coupled to the seat base.
10. The seat back of claim 1, wherein the frame is comprised of a plurality of tubular members fixedly attached to ends of each other.
11. A vehicle assembly comprising: a vehicle cabin; and a row of seats arranged within the vehicle cabin, the row of seats comprising at least one seat back comprising: a frame rotatably coupled at one end to a seat base, wherein the frame at least partially defines an opening in the seat back; and left and right panels rotatably coupled to respective sides of the frame.
12. The vehicle assembly of claim 11, wherein the row of seats comprises three seats.

- 13.** The vehicle assembly of claim 11, wherein the at least one seat back comprises a seat belt retractor coupled to the frame.
- 14.** The vehicle assembly of claim 11, wherein: the frame is rotatably coupled to the seat base by a hinge assembly; and the hinge assembly is configured to enable the frame to rotate between an upright position and a folded position where the frame is arranged horizontally above the seat base.
- 15.** The vehicle assembly of claim 11, wherein when the left and right panels are in a closed position, the left and right panels cover the opening.
- 16.** The vehicle assembly of claim 11, wherein when the seat back is arranged in an upright orientation and when the left and right panels are rotated outward about respective sides of the frame, the opening is exposed.
- 17.** The vehicle assembly of claim 11, wherein when the seat back is in a folded down position and when the left and right panels are rotated outward to a first position, the left and right panels form respective armrests.
- 18.** The vehicle assembly of claim 11, wherein when the seat back is in a folded down position and when the left and right panels are rotated outward, the opening is capable of receiving a storage bin that is removably coupled to a perimeter of the opening.
- 19.** The vehicle assembly of claim 11, wherein when the seat back is in a folded down position and when the left and right panels are rotated outward to a second position, the left and right panels are horizontally arranged and form respective work surfaces.
- 20.** A method of transforming a seat back, the method comprising: rotatably articulating a top end of a frame of the seat back from an upright orientation to a folded down orientation via a hinge assembly at a bottom end of the frame; and rotating left and right panels about respective sides of the frame to expose an opening at least partially defined by the frame.
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